Predicting Stellar Parameters and Detecting Anomalous Stars in the MaStar Stellar Spectral Library

SIADS 696 Milestone II - June 2026

Kevin Bao | Bonseok Koo | Zach Snow

1. Introduction

The MaNGA Stellar Library (MaStar) contains over 59,000 spectra from more than 24,000 unique stars observed with the Sloan Foundation Telescope between 2014 and 2020. Stellar spectra encode physical properties of stars directly in their absorption line patterns, making them a rich source of information for both supervised regression and unsupervised structure discovery.

This project pursues two goals. First, we train regression models to predict three continuous stellar parameters directly from observed spectra: effective temperature (Teff), surface gravity (log g), and metallicity ([Fe/H]). Second, we apply unsupervised clustering and anomaly detection to the full unlabeled catalog to discover hidden stellar populations and flag scientifically unusual objects.

Our best supervised model, XGBoost trained on continuum-normalized spectra without PCA, achieved a cross-validated R2 of 0.962 for Teff, 0.890 for log g, and 0.882 for [Fe/H], corresponding to MAEs of 128 K, 0.265 dex, and 0.166 dex respectively. Unsupervised clustering found a clear temperature-driven division between warm and cool stellar populations, consistent with the known spectral sequence. Anomaly detection flagged 760 candidate outliers via Isolation Forest, of which 263 were also flagged by LOF, representing the highest-confidence set of genuinely unusual stars.

2. Related Work

We identified four existing projects most closely related to our approach:

Yan et al. 2019 - MaStar Stellar Parameter Catalog

The MaStar team produced their own stellar parameter estimates using template matching and full-spectrum fitting against theoretical stellar models. This is the source of our INPUT_TEFF, INPUT_LOGG, and INPUT_FE_H labels. Unlike their template-matching approach, we apply supervised regression models directly to the observed spectra and add an unsupervised anomaly detection component that their pipeline does not include.

Fabbro et al. 2018 - StarNet

StarNet is a convolutional neural network trained directly on APOGEE DR13 spectra to predict Teff, log g, and [Fe/H], achieving precision comparable to the official APOGEE pipeline. This is the closest analogue to our supervised approach, but differs in two key ways: it applies a deep learning architecture to near-infrared APOGEE spectra at much higher resolution (R~22,500), while we use tree-based regression on the optical MaStar wavelength range (R~1800). StarNet also requires a large, high-S/N training set to be effective; our approach with XGBoost was specifically chosen because it is more robust on smaller labeled datasets like MaStar's ~3,300 labeled stars.

Anders et al. 2023 - XGBoost on Gaia XP spectra

Anders et al. benchmarked several supervised learning methods for stellar parameter estimation from Gaia DR3 XP spectra and found that XGBoost outperformed neural network architectures for this tabular spectral data, producing competitive Teff, log g, and metallicity estimates for millions of stars. This directly validates our model choice. The key difference is that Gaia XP spectra are delivered as internally-calibrated spectral coefficients rather than flux arrays, making them naturally suited to tabular models. Our work applies the same reasoning to continuum-normalized MaStar flux arrays and includes an unsupervised anomaly detection component that their catalog does not.

Reis et al. 2018 - Outlier detection in APOGEE

Reis et al. applied an unsupervised random forest similarity measure to APOGEE stellar spectra, using it for clustering and outlier detection, successfully recovering previously unknown Be-type stars, spectroscopic binaries, and other rare objects that could not be found through standard parameter fitting alone. This is the most direct precedent for our unsupervised component. Our approach differs in that we use PCA followed by K-Means, HDBSCAN, and Isolation Forest rather than an unsupervised random forest similarity measure, and we explicitly compare the overlap between two anomaly detectors (Isolation Forest and LOF) to identify the highest-confidence anomaly candidates.

3. Data Source

We use the MaStar Good Spectra and Summary Catalog from SDSS Data Release 17, available at https://www.sdss4.org/dr17/mastar/mastar-data-access/. The dataset is distributed as FITS files. Key tables include:

- mastarall: summary catalog with per-star metadata and stellar parameter labels (INPUT_TEFF, INPUT_LOGG, INPUT_FE_H, MJDQUAL, S2N, BADPIXFRAC, MANGAID)
- GOODVISITS: visit-level table used for best-visit selection via S/N ranking (S2N column)
- goodspec: matched spectral flux arrays on a shared wavelength grid covering 3,622 to 10,354 Angstroms at R~1800 resolution (~4,500 wavelength bins per spectrum)

The full catalog contains 59,000+ spectra from 24,130 unique stars. After quality filtering and selecting the labeled subset with valid stellar parameters, our supervised working set contains 3,291 unique stars. The full unsupervised set includes approximately 24,000 unique stars passing quality cuts. The dataset has no time-series element; each spectrum is a single-epoch observation (or best-visit selection from multiple visits).

4. Feature Engineering

4.1 Quality Filtering

Stars were filtered using MJDQUAL (master quality bitmask), S2N (signal-to-noise ratio), and BADPIXFRAC (fraction of bad pixels) to remove unreliable spectra before modeling.

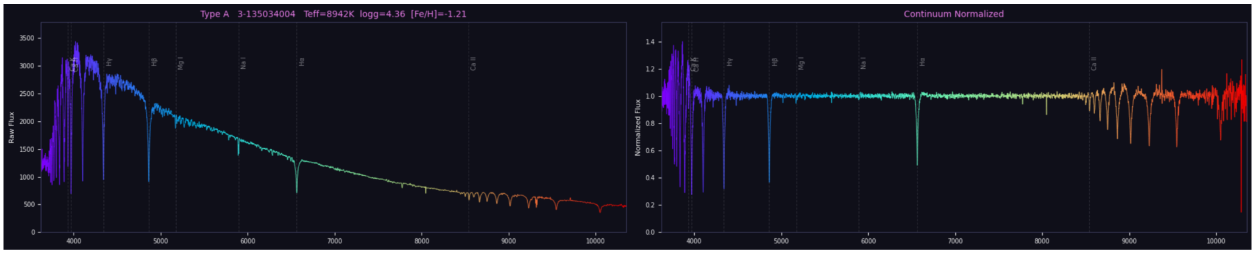
4.2 Best-Visit Selection

For stars with multiple visits, the best visit was selected by ranking on S2N from the GOODVISITS catalog table, retaining the highest signal-to-noise observation per unique star.

4.3 Continuum Normalization

Spectra were continuum-normalized using a scipy median filter with a 101-bin window. This approach proved robust across spectral types, removing overall flux level differences so models train on spectral shape rather than absolute flux.

Figure 4.3.1: *Raw vs. continuum-normalized flux for a representative MaStar spectrum*



4.4 Final Features

The final feature set for both supervised and unsupervised models consists of 4,614 continuum-normalized flux values per spectrum, covering 3,622 to 10,354 Angstroms on the shared MaStar wavelength grid. For unsupervised learning, PCA was applied after standardizing features to zero mean and unit variance. The full wavelength grid is listed in Appendix A.

Part A: Supervised Learning

For the supervised learning portion, we started from the final continuum-normalized feature dataset described in Section 4. Each row represented one labeled star, and each column represented a wavelength-bin flux value on the shared MaStar wavelength grid. The supervised dataset contained 3,291 labeled stars with valid values for effective temperature (Teff), surface gravity (log g), and metallicity ([Fe/H]).

Our supervised task was to predict these three stellar parameters from the continuum-normalized flux features. Since Teff, log g, and [Fe/H] have different physical meanings, units, and error scales, we trained and evaluated separate regression models for each target rather than combining the three targets into one score.

Before training models, we checked the distribution of spectral types in the labeled dataset. The data was heavily imbalanced: K-type stars made up 58.8% of the sample, followed by G (19.3%), F (14.3%), A (5.4%), and M (2.2%). To avoid a train/test split that over- or under-represented rare spectral types, we used an 80/20 stratified split, producing 2,632 training stars and 659 test stars. The stratification was effective: train and test distributions matched the full dataset to within 0.1 percentage points across all five spectral types (see Appendix B, Figure B1 for the full distribution comparison).

We then compared three supervised model families.

- Ridge Regression: a regularized linear baseline used to establish a simple performance floor.
- Random Forest Regressor: a nonlinear tree-based model used to capture interaction between wavelength regions.
- XGBoost Regressor: a gradient-boosted tree model used as a stronger nonlinear model for high dimensional tabular spectra.

For each model family, we compared PCA and non-PCA versions. This comparison helped us test whether dimensionality reduction improved prediction performance or whether the original wavelength features should be preserved. PCA was expected to help Ridge regression because Ridge is a linear model and the wavelength features are high-dimensional and highly correlated. For Random Forest and XGBoost, PCA was treated as an ablation test because tree-based models may benefit from using original wavelength-level information directly.

5.1 Hyperparameter Tuning

We used a step-by-step tuning strategy rather than one large exhaustive grid search. First, we compared the three model families with and without PCA. This allowed us to evaluate both model choice and feature representation at the same time.

For Ridge Regression, the main tuning decision was whether to use original wavelength features or PCA-reduced features. Ridge without PCA performed poorly, while Ridge with PCA improved significantly on the test set. This confirmed that dimensionality reduction was useful for the linear baseline.

For Random Forest and XGBoost, PCA was tested as an ablation rather than assumed to be necessary. The result showed that the XGBoost model performed best without PCA amongst all the other models, suggesting that the original continuum-normalized wavelength features contained spectral information that was useful for prediction.

After XGBoost without PCA was confirmed to be the strongest model, we performed sensitivity analysis on two key XGBoost hyperparameters: the number of estimators (n_estimators) and maximum tree depth (max_depth). The number of estimator analysis tested values from 50 to 900 trees, while maximum tree depth analysis tested from 2 to 8. These experiments helped us determine whether the final model was underfitting, overfitting or already near a stable performance range.

Finally, we used 5-fold stratified cross-validation to evaluate model stability. Stratification was based on spectral type so that each fold preserved the overall distribution of stellar types as much as possible.

6. Supervised Evaluation

6.1 Metrics

We evaluated each model using both R2 and Mean Absolute Error (MAE). R2 was the main comparison metric because it measures how well the model's predictions matched the actual target values. Since the goal was to predict continuous stellar parameters from spectra, R2 helped us evaluate whether each model was learning the overall relationship between the spectral features and the true stellar labels.

MAE was also reported because it is easier to interpret in the physical units of each target. For example, a Teff MAE of 128 means that the model's temperature prediction was off by 128 Kelvin on average. The same rules apply to our predictions for gravity and metallicity. Therefore, R2 was used mainly to compare model fit and predictive strength, while MAE was used to understand the size of the prediction error.

6.2 Overall Results

We first compared model performance using the 80/20 stratified train/test split. The test set contained 659 stars that were not used during model training, so it gave us an initial estimate of how well each model generalized to unseen spectra.

The overall pattern was clear from the R2 scores. Ridge Regression without PCA performed poorly, with low or negative R2 values across the three target variables. This suggested that the original wavelength feature space was too high-dimensional and correlated for a simple linear model to handle effectively. After PCA was applied, Ridge Regression improved significantly, especially for Teff. This showed that dimensionality reduction helped the linear baseline by compressing the wavelength features into a smaller set of principal components.

However, the strongest overall model was XGBoost without PCA. On the test set, XGBoost without PCA achieved R2 values of 0.967 for Teff, 0.887 for log g, and 0.864 for [Fe/H]. These results showed that XGBoost was able to learn strong nonlinear relationships between the continuum-normalized wavelength features or shape of spectra and the true stellar parameters.

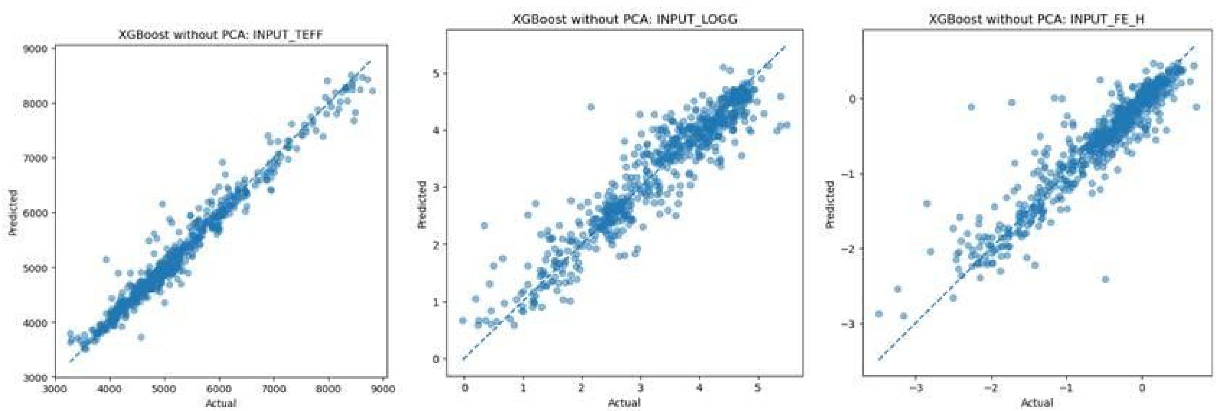
Table 1 summarizes the holdout test-set performance for each model and target.

Table 6.2.1: *Holdout test-set performance by model and stellar parameter*

	Model	Target	MAE	R2
11	RandomForest with PCA	INPUT_FE_H	0.253870	0.759939
8	RandomForest without PCA	INPUT_FE_H	0.323940	0.653789
5	Ridge with PCA	INPUT_FE_H	0.228188	0.738636
2	Ridge without PCA	INPUT_FE_H	0.501452	-0.493322
17	XGBoost with PCA	INPUT_FE_H	0.192674	0.828418
16	XGBoost without PCA	INPUT_FE_H	0.170777	0.864083
10	RandomForest with PCA	INPUT_LOGG	0.451894	0.711696
7	RandomForest without PCA	INPUT_LOGG	0.467603	0.727666
4	Ridge with PCA	INPUT_LOGG	0.327134	0.788546
1	Ridge without PCA	INPUT_LOGG	0.793714	-0.441123
15	XGBoost with PCA	INPUT_LOGG	0.316980	0.822817
14	XGBoost without PCA	INPUT_LOGG	0.268047	0.887364
9	RandomForest with PCA	INPUT_TEFF	142.917063	0.961398
6	RandomForest without PCA	INPUT_TEFF	128.483515	0.967524
3	Ridge with PCA	INPUT_TEFF	175.635516	0.931074
0	Ridge without PCA	INPUT_TEFF	478.974201	0.063434
13	XGBoost with PCA	INPUT_TEFF	146.813524	0.960580
12	XGBoost without PCA	INPUT_TEFF	128.543452	0.966865

The R2 comparison explains why XGBoost without PCA was selected as the final supervised model. Random Forest without PCA had a slightly higher R2 for Teff, but XGBoost without PCA performed better overall across log g and [Fe/H], which were harder targets. XGBoost also had the lowest MAE for log g and [Fe/H], making it the strongest balanced model across all three stellar parameters.

Figure 6.2.1: *XGBoost Actual vs. Predicted*



The predicted vs. actual plots supported the numerical results. Teff predictions followed the one-to-one line most closely, which is consistent with the high R2 score. The log g and [Fe/H] plots showed more spread but still showed clear predictive structure. This suggests that the model learned useful spectral patterns for all three parameters, while temperature remained the easiest target to predict.

After the holdout test-set comparison, we also used 5-fold stratified cross-validation to check whether the model ranking was stable across different data splits. This cross validation was run separately on the full labeled dataset using spectral type as the stratification label. The results showed that Random Forest and XGBoost had much smaller fold to fold variation than Ridge with PCA. Ridge with PCA looked strong on the original test split, but its cross-validation results were unstable, suggesting that it was sensitive to which stars appeared in each fold.

Table 6.2.2: *Cross-validated R2 and MAE by model and stellar parameter, mean ± standard deviation across 5 folds*

	Model	Target	Mean_R2	Std_R2	Mean_MAE	Std_MAE
0	Ridge with PCA	INPUT_TEFF	-541.253537	1014.096981	713.616850	852.241087
1	Ridge with PCA	INPUT_LOGG	-1320.379270	2589.396521	1.188199	1.514065
2	Ridge with PCA	INPUT_FE_H	-10369.681459	20739.794089	1.531140	2.609012
3	RandomForest without PCA	INPUT_TEFF	0.962017	0.006298	129.061871	6.011611
4	RandomForest without PCA	INPUT_LOGG	0.877079	0.028330	0.269329	0.012236
5	RandomForest without PCA	INPUT_FE_H	0.869625	0.018123	0.173813	0.005488
6	XGBoost without PCA	INPUT_TEFF	0.962140	0.006190	128.412123	5.644567
7	XGBoost without PCA	INPUT_LOGG	0.889520	0.013666	0.264981	0.009805
8	XGBoost without PCA	INPUT_FE_H	0.881720	0.015609	0.165621	0.002862

Overall, the holdout test-set R2 results showed that XGBoost without PCA had the strongest predictive fit across the three targets, while the cross-validation MAE results confirmed that its performance was stable across folds.

6.3 Feature Importance and Ablation

After selecting XGBoost without PCA as the final model, we examined feature importance to understand which wavelength regions contributed most to the prediction. Because the final model did not use PCA, the feature importance values mapped directly back to the original wavelength bins.

For Teff, the highest-importance features correspond to the hydrogen Balmer series (Hα at 6563 Å, Hβ near 4861 Å, Hγ near 4340 Å), consistent with Balmer absorption being the primary temperature diagnostic across our spectral type range. For log g, top features concentrate around Ca II K (3934 Å) and the Mg I b triplet (5164-5252 Å), both well-established pressure-broadening diagnostics that distinguish giants from dwarfs. For [Fe/H], the dominant feature at 5891 Å corresponds to the Na I D doublet, with remaining top features falling in Fe I blend and Mg I regions across 4800-5200 Å. Full top-20 wavelength tables for each target are provided in Appendix B.

We also used the PCA and non-PCA model comparison as an ablation test. PCA helped Ridge Regression significantly, but it did not improve the final XGBoost model. XGBoost performed best when using the original continuum normalized wavelength features.

The ablation results showed that PCA was useful for stabilizing a linear model, while nonlinear model performed best with original wavelength representation. PCA reduced the number of features, but it also mixed the original wavelengths together.

Table 6.3.1: *Ablation result*

	Model	Target	MAE	R2
11	RandomForest with PCA	INPUT_FE_H	0.253870	0.759939
8	RandomForest without PCA	INPUT_FE_H	0.323940	0.653789
5	Ridge with PCA	INPUT_FE_H	0.228188	0.738636
2	Ridge without PCA	INPUT_FE_H	0.501452	-0.493322
17	XGBoost with PCA	INPUT_FE_H	0.192674	0.828418
16	XGBoost without PCA	INPUT_FE_H	0.170777	0.864083
10	RandomForest with PCA	INPUT_LOGG	0.451894	0.711696
7	RandomForest without PCA	INPUT_LOGG	0.467603	0.727666
4	Ridge with PCA	INPUT_LOGG	0.327134	0.788546
1	Ridge without PCA	INPUT_LOGG	0.793714	-0.441123
15	XGBoost with PCA	INPUT_LOGG	0.316980	0.822817
14	XGBoost without PCA	INPUT_LOGG	0.268047	0.887364
9	RandomForest with PCA	INPUT_TEFF	142.917063	0.961398
6	RandomForest without PCA	INPUT_TEFF	128.483515	0.967524
3	Ridge with PCA	INPUT_TEFF	175.635516	0.931074
0	Ridge without PCA	INPUT_TEFF	478.974201	0.063434
13	XGBoost with PCA	INPUT_TEFF	146.813524	0.960580
12	XGBoost without PCA	INPUT_TEFF	128.543452	0.966865

6.4 Sensitivity Analysis

After selecting XGBoost without PCA as the best performing supervised model, we tested how sensitive its performance was to two key hyperparameters: the number of estimators (n_estimators) and maximum tree depth (max_depth). The purpose was to check whether the model was still improving or whether performance had already reached a stable range.

For the number of estimators, we tested values from 50 to 900 while keeping the other XGBoost settings fixed. The largest improvement occurred between 50 and 150 trees. For example, Teff R2 improved from 0.953 at 50 trees to 0.966 at 150 trees, while Teff MAE decreased from 169.35 K to 129.78 K. After 150 to 300 trees, the improvement became much smaller, suggesting that the model had already captured most of the useful predictive signal.

Full sensitivity curves for all three targets across both hyperparameters are shown in Appendix B. For Teff, these curves illustrate the rapid early improvement and subsequent plateau clearly.

We also tested max depth values from 2 to 8. Depth 2 performed slightly worse, suggesting that very shallow trees were not complex enough to capture spectral relationships. Performance improved around depths 3 to 5, then stopped improving at deeper values. Depth 5 gave the lowest MAE in this test, but the final model used depth 3 as a more conservative setting that still performed strongly while reducing overfitting risk.

Overall, the sensitivity analysis showed that the XGBoost model was relatively stable after moderate model setting. Increasing the number of estimators beyond 300 or increasing tree depth beyond 5 did not produce meaningful additional information.

6.5 Tradeoffs

The supervised results showed several tradeoffs. Ridge Regression was the simplest model and improved significantly with PCA, but it was not stable in cross validation. Random Forest and XGBoost were more complex, but they captured nonlinear spectral relationships better and produced stronger results.

There was also a trade off between dimensionality reduction and preserving wavelength information. PCA helped Ridge because it reduced high-dimensional wavelength space into fewer components. However, nonlinear models performed best without PCA, suggesting that the original wavelength-level features contained useful local spectral information that PCA may have diluted together.

Another tradeoff was model complexity versus overfitting risk. The sensitivity analysis showed that deeper trees improved performance up to a point, but depth values beyond 5 did not continue improving the model. This supported the use of conservative final depth setting of 3.

6.6 Failure Analysis

For failure analysis, we used the final XGBoost without PCA model and identified test-set stars with the largest prediction errors. Since, Teff, log g and [Fe/H] have different units, we scaled each target's absolute error by training-set standard deviation and summed the scaled error. This allowed us to compare overall failure severity across the three targets.

Table 6.6.1: *Top failed prediction cases from final XGBoost model*

	Original_Index	MANGAID	SPECTRAL_TYPE	Worst_Target	Total_Scaled_Error
626	2257	7-17129068	K	INPUT_FE_H	5.606003
606	955	3-151550654	K	INPUT_FE_H	4.043756
67	960	3-151672729	K	INPUT_FE_H	3.045430
610	1783	60-2051287173829740032	M	INPUT_LOGG	3.032123
104	3060	7-7079819	K	INPUT_TEFF	2.935028
538	684	3-139937884	K	INPUT_FE_H	2.856939
442	2434	7-18053524	K	INPUT_FE_H	2.754945
13	766	3-142072281	K	INPUT_TEFF	2.440749
131	2000	7-11077539	G	INPUT_FE_H	2.305933
581	1759	60-1995012897413744384	A	INPUT_FE_H	2.193289

The largest failures were mostly related to unusual parameter values rather than random error. Several of the top failed records were very metal poor K type stars where the model predicted metallicity closer to the common training range. Another major failure was an M type star where the model overpredicted log g. This is consistent with the data imbalance as M type stars were rare in the labeled sample.

When errors were grouped by spectral type, M type stars had the highest average scaled error, followed by A types stars. This suggests that future improvement should focus on rare or underrepresented cases. Adding more M type stars and A type stars and metal poor stars would likely improve performance.

Table 6.6.2: *Error summary by spectral type*

	Count	Mean_Total_Scaled_Error	Median_Total_Scaled_Error
SPECTRAL_TYPE			
M	14	0.981692	0.654359
A	36	0.820768	0.681023
G	127	0.606055	0.501659
F	94	0.579667	0.499863
K	388	0.529280	0.370394

Part B: Unsupervised Learning

7. Methods

Unsupervised learning used the full ~24,000 star visit catalog, after quality filtering, and without any stellar parameter labels. We first took the best visit per star and removed any data quality issues regarding zero bins, imputed median values where prudent, cut out any bins without at least 95% coverage, and then applied standard scaling and continuum normalization.

Figure 7.1: *Zero-bin counts per star (left), wavelength coverage at 99% and 95% thresholds (center), and red-end coverage drop-off near 10,200 Å (right).*

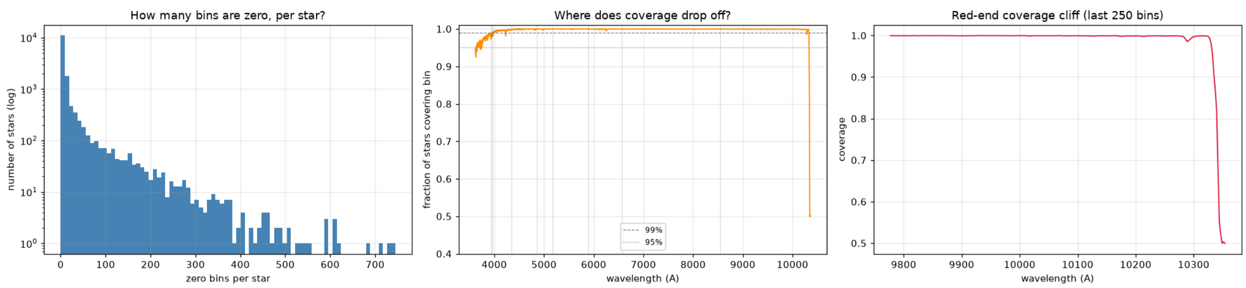
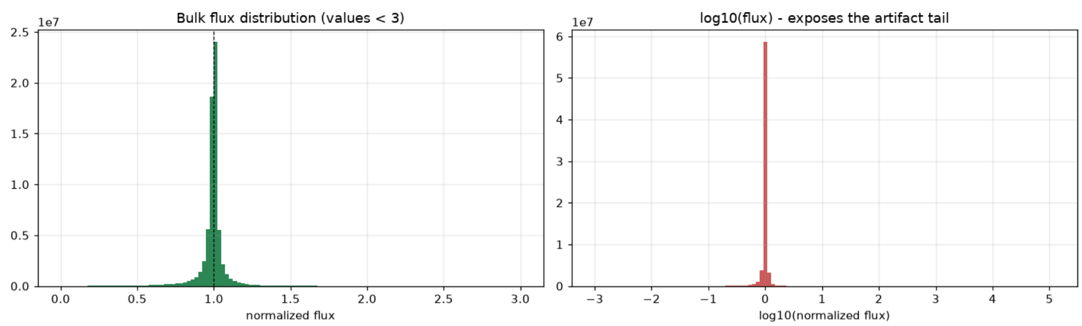
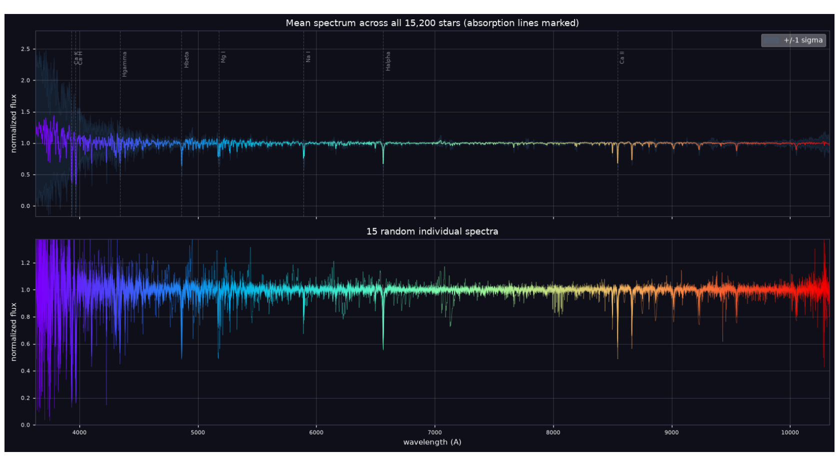


Figure 7.2: *Continuum-normalized flux distribution in bulk (left) and log scale (right), showing tight concentration near 1.0 with a small artifact tail.*



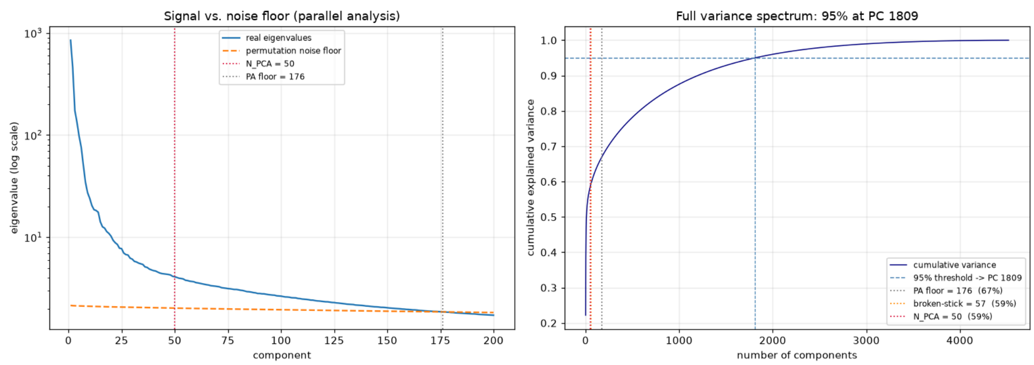
This left us with 15,200 star visits that were cleaned and viable for unsupervised modeling.

Figure 7.3: *Mean continuum-normalized spectrum with ±1σ band and labeled absorption lines (top), and 15 randomly sampled individual spectra (bottom).*



We now had 4520 continuum normalized flux values in our dataset, which is too many dimensions to cluster directly, so we used reduced dimensionality using Principal Component Analysis (PCA). We determined the optimal number of dimensions to use for our following steps by using a scree plot as an initial gauge, and then a parallel analysis of the signal to noise floor and a broken stick analysis against the initial 95% variance we had proposed.

Figure 7.4: *Parallel analysis scree plot with noise floor at PC 176 and chosen N_PCA=50 marked (left), and cumulative explained variance with broken-stick (PC 57) and 95% threshold (PC 1809) reference lines (right).*

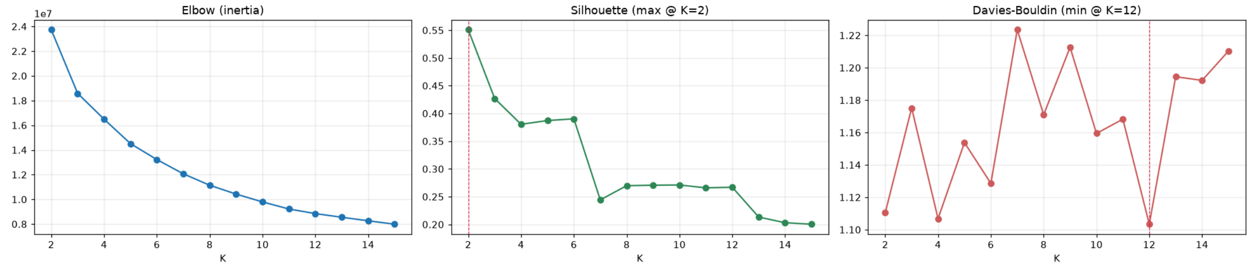


Through this, we ended up using N_PCA = 50 as our path forward because the noise floor was found to be at 176, and the broken stick was found to be at 57 components. By choosing a value slightly before the broken stick, this helped provide us with a clear consistent signal path for all the unsupervised learning analysis' moving forward. If we were to use our initially proposed guide of 95% variance-explained, it would've taken 1809 components, which was well past the noise floor. 50 components explained about 59% of the variance, which was good enough to provide our analysis with clean and structured clusters. Using this, we moved into our clustering steps of our analysis.
For clustering, we utilized two methods.

- K-Means (post-PCA): a fast, interpretable centroid-based baseline that splits the data into round clusters. The number of clusters k was selected by comparing the value outputs of the silhouette score, the inertia elbow, and the Davies-Bouldin score.
- HDBSCAN (post-PCA): a variable density-based hierarchical tree method that does not require specifying k in advance and can identify irregular cluster shapes, with outlier points left unclustered. More computationally complex but great for noisy or varying density datasets like this one.

We started with K-means clustering, and iterated through a k hyperparameter value range of 2 to 15 and checked the findings. Even though the Davies-Bouldin score had the lowest values around K=12, the surrounding K values were relatively high in comparison which made us question the variability and accuracy of choosing K=12. However, using the silhouette score, we say that at K=2, we saw the best balance of a high silhouette score and a relatively low Davies Bouldin score, so we used K=2. We also used a K=6 value later on as a comparison against our labeled data to interpret the cluster effectiveness, however note that we were careful to not use any of the labeled data in our modeling as this would have been leakage.

Figure 7.5: *K-means cluster selection metrics across k=2 to 14: inertia elbow (left), silhouette score with maximum at K=2 (center), and Davies-Bouldin index with minimum at K=12 (right).*



We then moved on to using the tree-based HDBSCAN to see if this approach would provide us with a different result. The HDBSCAN hyperparameter sweep consistently produced 2 clusters with approximately 35% of stars labeled as noise across all tested combinations of min_cluster_size and min_samples (full sweep results in Appendix B, Figure B4). The low DBCV scores across the grid confirmed that the data has no strong density-separated structure, which is consistent with stars living on a continuous temperature sequence rather than forming discrete populations.

Ablation tests varying N_PCA across 10, 30, 50, and 100 dimensions all returned K=2 with ARI=1.0 relative to the N_PCA=50 solution, confirming that the two-cluster result is stable and not an artifact of our dimensionality choice. Dropping the blue Ca H&K block below 4200 Å also had minimal impact (ARI=0.981 vs. the full feature set), suggesting the clustering signal is distributed broadly across the spectrum rather than driven by any single wavelength region (full ablation results in Appendix B, Table B5).

The K=2 clusters map cleanly onto a temperature-driven division visible in the Kiel diagram: a large warm-star cluster and a smaller cool-star group. At K=6 the clustering recovers a more granular but recognizable pattern consistent with hot A-type, G/K main-sequence, and cool M-type populations, confirming the model is responding to genuine spectral structure. Kiel diagrams for both K=2 and K=6 are shown in Appendix B, Figures B6 and B7. Label data was not used at any point in the modeling; the K=6 comparison was purely a sanity check.

Anomaly Detection

Next, we moved on to the anomaly detection portion of our unsupervised learning model. We used both Isolation Forest and Local Outlier Factor (LOF).

- Isolation Forest: A tree-based and efficient method of anomaly detection that uses repeated random splits of the data to see how each datapoint (star) gets cut off from its peers.
- Local Outlier Factor (LOF): A local density-based method of anomaly detection that uses distance from a star and its neighbors against the neighbor's relative neighbors.

While LOF was initially proposed, Isolation Forest was used as our primary detector since our peer review comments called out how LOF can cause issues with PCA-reduced spectra. Isolation Forest helped solve this problem, however we still wanted to utilize LOF as a comparison and used the overlap of both methodologies as stars that are the most likely to be anomalous.

In Isolation Forest, we initially used N_PCA = 50 dims as per our initial methodology findings, ran a contamination hyperparameter sweep at a tested optimum of 0.05, and flagged 760 (5% of) stars as anomalous.

Figure 7.6: *Isolation Forest anomaly score distribution with 5% contamination threshold marked (left), and anomaly score vs. zero-bin count per star on a log scale (right).*

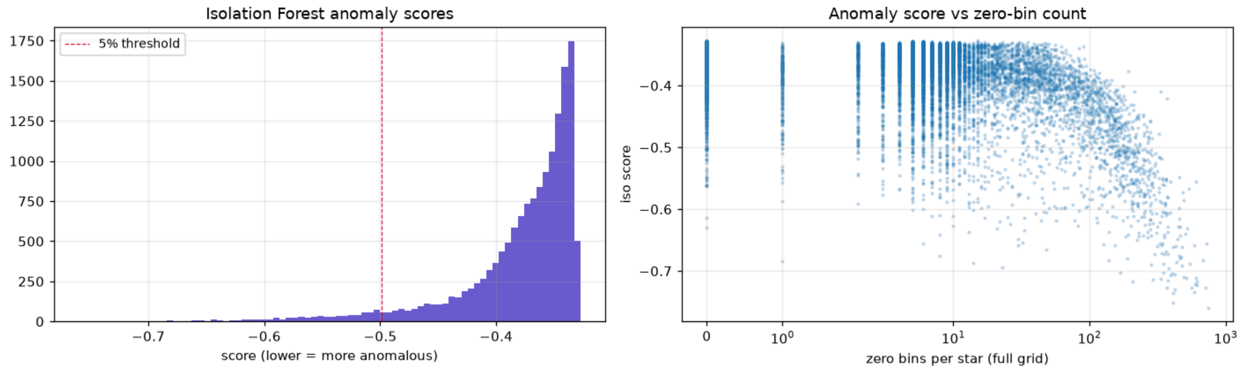


Table 7.1: *Isolation Forest contamination sensitivity analysis showing flagged star counts*

```
Baselines -> A/M types among labeled: 0.076 | high-zero-count among all stars: 0.050

Contamination sensitivity:
 contamination  n_flagged  n_labeled_flagged  frac_AM_among_labeled  frac_high_zero_count
          0.02        304                 30                  0.300                 0.632
          0.05        760                 94                  0.340                 0.495
          0.10       1520                235                  0.281                 0.367
```

After cross-referencing against LOF, we found that there was an overlap of 35% of IF flagged stars also flagged by LOF. This 35% or 263 stars are the highest likelihood of being actually anomalies.

However, at this point we also wanted to test to see if our N_PCA dimensionality was affecting our anomaly detection model as well, so we did an ablation test here as well. By using fields like frac_AM_among_labeled and frac_high_zero_count from the labeled datasets, we could see how PCA dimensionality effects our Isolation Forest model and tested the hyperparameter range of pca_dims from 5-176 (our noise floor). However, even though this resulted in showing us the task optimal point would have been N_PCA=15, we still kept it at N_PCA=50 because labeled data was used to determine this value, and in a true unsupervised learning model this data wouldn't have been available for us to reference against, thus invalidating our finding.

Finally, we generated some visualizations of our results, a brief failure analysis, and called out some potential limitations we had as we were undergoing this portion of our project.

8. Unsupervised Evaluation

8.1 Metrics

For clustering, we report silhouette score, Davies-Bouldin Index, and inertia (for K-Means) to assess clustering quality without labels. As a sanity check, we examined whether unsupervised clusters align with known spectral types in the labeled subset. For anomaly detection we compared two different anomaly detection methodologies and utilized the overlap between both methods as our flagged anomalies. We also used the same sanity check to see if our anomalies in A and M spectral types compare against the labeled subset as a baseline again too. In addition, we also tried to pull out which flagged stars are truly anomalous stars or just data-quality issues with empty wavelength bins.

8.2 Overall Results

Clustering: Our model essentially splits the stars into a large warm cluster (cluster 0: 13,274 stars, 87.3%) and a smaller cool star group (cluster 1: 1,926 stars, 12.7%) at K=2. After comparing against the labeled data, cluster 0 essentially holds all the A, F, G, and some K type stars, whereas the M and K type stars are in cluster 1. This primarily gives us a clean temperature-driven division. HDBSCAN showed us almost the same results however with a separated ~35% of stars called out as noise. This isn't necessarily a bad thing, as this moreso shows what we know about the stars as a whole - stars sit on a continuous sequence rather than having very well-separated clean cut clusters.

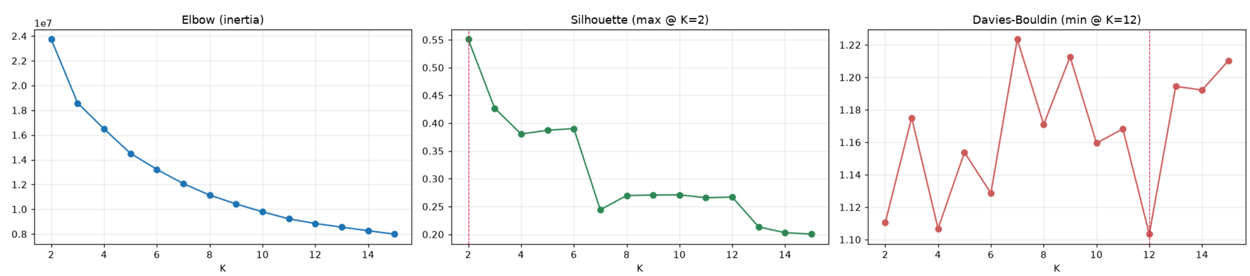
By breaking things out more into the k=6 clusters, we see a much more recognizable pattern of hot A-type stars, G/K main star types, and a cooler M-type cluster. This tells us that our model is working correctly, it's just that stars live on a spectrum so clustering follows that same pattern.

Table 8.2.1 summarizes clustering quality metrics for both methods.

Table 8.2.1: *Clustering quality metrics by method*

Method	Silhouette	Davies-Bouldin	Notes
k-means, k=2	0.551	1.110	2 clusters (87.3% / 12.7%)
k-means, k=6	0.390	1.129	interpretation view
HDBSCAN	n/a (noise labels)	DBCV ~0.0001	2 clusters, 34.6% noise

Figure 8.2.1: *K-means cluster selection metrics across k=2 to 15: inertia elbow (left), silhouette score with maximum at K=2 (center), and Davies-Bouldin index with minimum at K=12 (right).*



Anomaly Detection: a 5% contamination Isolation Forest flagged 760 stars from the dataset as potentially anomalous. After comparing against the labeled dataset as a baseline, we found that 35% of the flagged stars were A and M type stars, which are 7.6% of the overall labeled dataset. This tells us that the anomaly detection is good at catching rare hot and cool stars. But 49% of the flags are also due to a high zero count spectra with empty wavelength bins. This tells us the model is also good at catching data quality issues well. Much better than LOF's 20%. But of the two models utilized, there were 263 stars (35% of the Isolation Forest flagged stars) that were overlapping between both models. These are the highest likelihood of being truly anomalous stars.

Figure 8.2.2: *UMAP projection of the 15,200-star catalog colored by K-means cluster assignment at K=2 (left) and Isolation Forest anomaly score (right).*

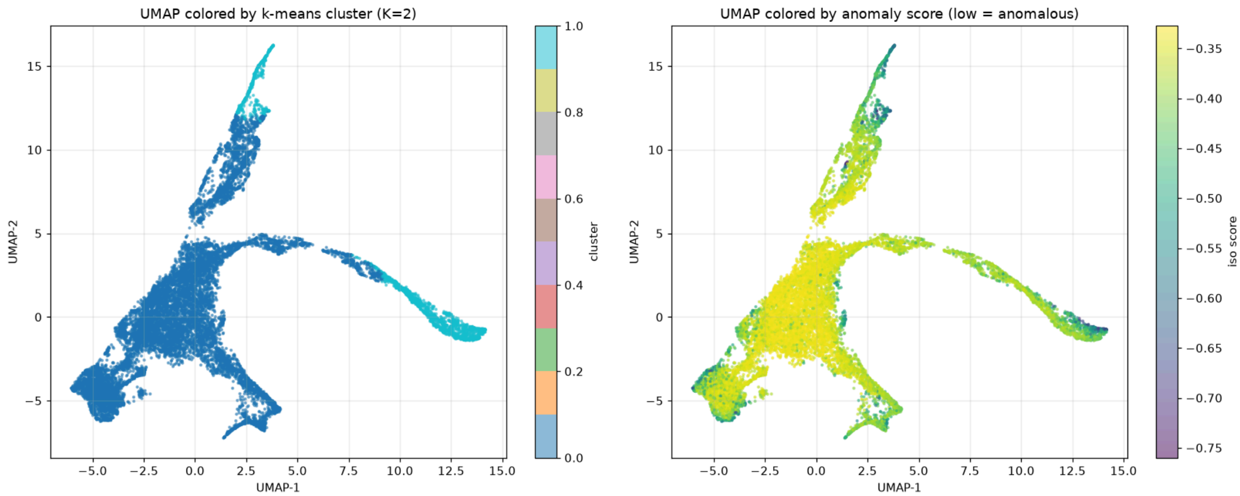
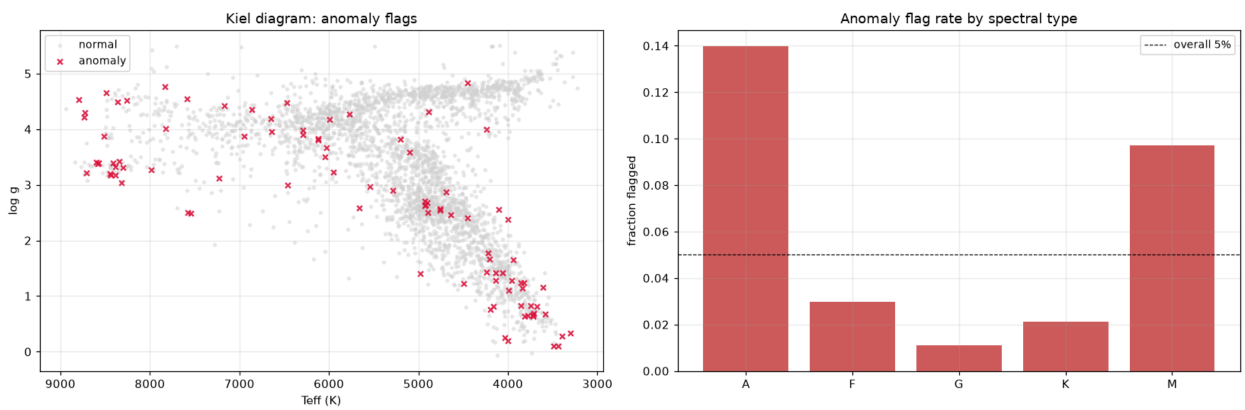


Figure 8.2.3: *Anomaly flags overlaid on the Kiel diagram (left) and anomaly flag rate by spectral type relative to the overall 5% contamination threshold (right).*



Representative spectra from the highest-ranked Isolation Forest anomalies are shown in Appendix B, Figure B8.

8.3 Sensitivity Analysis

We ran a sensitivity analysis of multiple parts of our analysis, including testing hyperparameters of N_PCA, K clusters, and contamination to see how our results might differ by adjusting these values, processes, or features. But as an example, for the contamination portion of our Isolation Forest anomaly detection model, we adjusted the contamination rate to see how that would affect the number of stars our model flagged as anomalies. By dropping from 5% down to 2%, the model then flags the 304 most extreme outlier stars and pushes the data quality share up to 63%. This tells us that the most extreme outliers are mostly data quality/broken spectra issues. By raising the contamination up to 10%, this flags 1520 stars, calling out data quality issues down to 37% and rare star type flagged anomalies at 28%. The 5% setting was a reasonable middle ground for us.

Table 8.3.1: *Isolation Forest contamination sensitivity analysis. Baselines: A/M types = 7.6% of labeled set, high-zero-count = 5.0% of full catalog.*

contamination	# flagged	# labeled flagged	A/M among labeled	high-zero-count share
0.02	304	30	0.300	0.632
0.05	760	94	0.340	0.495
0.10	1520	235	0.281	0.367

As mentioned before, we also did sensitivity analyses on PCA dimensionality in clustering at N_PCA = [10, 30, 50, 100] to test the K cluster split, but we found that the K=2 was stable across all values of N_PCA.

9. Discussion

9.1 Supervised Learning (Part A)

From the supervised regression results, we found that MaStar spectra contain strong information for predicting stellar parameters, especially effective temperature. This made sense because temperature strongly affects the overall spectral shape and major absorption features. XGBoost without PCA was the strongest overall model because it performed well across all three targets, including the harder targets like log g and [Fe/H].

One interesting result was that PCA helped Ridge Regression, but not XGBoost. Ridge benefited from PCA because it is a linear model working with thousands of correlated wavelength features. However, XGBoost performed better without PCA, suggesting that the original wavelength features contained useful local spectral information that the tree-based model could use directly.

The main challenges were class imbalance and high dimensionality. K-type and G-type stars made up most of the labeled dataset, while A-type and M-type stars were much less common. We used stratified train/test splits and stratified cross-validation to help preserve the spectral-type distribution. Even then, failure analysis showed that rare cases were harder for the model, especially M-type stars and very metal-poor K-type stars.

With more time, we could improve the supervised model by adding more labeled examples for rare spectral types and metal-poor stars, testing sample weighting, or trying neural network models such as one-dimensional CNNs. Cross-matching MaStar with Gaia could also add external features like photometry or distance-related information, which may help improve difficult predictions such as log g and [Fe/H].

9.2 Unsupervised Learning (Part B)

After analyzing the MaStar Stellar Spectra database in its full entirety in the unsupervised learning portion of the analysis, we found that stellar populations don’t cluster into very easily discernible clusters. Instead, the data shows that it’s basically one long temperature spectrum. While some members of our team were familiar with this idea, this was surprising to some of us as well, but it definitely makes sense. This was solidified through the data as both clustering methods (K-means clustering and HDBSCAN) showed a K=2 split independently of the sensitivity analysis that we conducted.

Anomaly detection worked to flag both rare hot and cool stars along with broken spectra scans, which can help to improve future data inputs if new scans were inducted into the dataset. It is still limited in separating out between the two (astrophysically interesting vs bad data) but it can at least flag these effectively as a whole.

We were challenged by the zero-padding issue at the red-end of the spectra as it initially caused issues in our analysis, particularly in the anomaly detection portion and it's scores. By going back and solving this issue, this helped us improve our findings. In addition, there were no O or B type stars in the dataset which could be a challenge out in a "real-world" application use, but we had to work around this in our analysis as we were validating those star types.

With some more time, we could maybe improve PCA for something like UMAP or an autoencoder to better capture any nonlinear manifold structures in our dataset or cross-reference our dataset against a larger external dataset like the Gaia dataset to help improve our anomaly classificaions. This would help us improve our model to tell data noise from real unique anomaly stars.

10. Ethical Considerations

10.1 Supervised Learning

The labeled subset of our data skews heavily towards cooler K-type and G-type stars, so models trained on it will tend to be more reliable in the 4,000-6,000K range than at the extremes. Hot stars and metal-poor halo stars are underrepresented in our data, and a model with little exposure to them during training can still return precise-looking predictions at inference time.

That's the core problem. Regression errors don't carry an obvious flag for this. If predictions are passed into downstream pipelines without uncertainty estimates, confidently wrong parameter values can propagate quietly into catalogs. Reporting calibrated uncertainty alongside point estimates, and explicitly flagging predictions that fall far outside the training distribution, would go a long way in preventing that.

10.2 Unsupervised Learning

The central ambiguity in anomaly detection is that "anomalous" can mean two very different things: a genuine astrophysical rarity, or a spectrum with instrumental problems like bad sky subtraction or zero-padded bins. This showed up concretely in our results, where roughly half of the stars flagged by Isolation Forest turned out to be high-zero-count spectra rather than unusual stars. An automated pipeline acting directly on those flags risks cutting scientifically interesting objects while elevating broken spectra for otherwise ordinary stars.

Clustering also introduces reproducibility concerns. The structure that emerges from PCA-reduced spectra depends on choices that are not always physically motivated, such as how many components to retain or what min_cluster_size to use for HDBSCAN. Different reasonable choices can produce meaningfully different groupings, so any conclusions drawn from cluster structure should document those decisions clearly and include sensitivity analysis, as we have done here.

11. Statement of Work

The following table describes the contribution of each team member to this project.

Team Member	Contributions
Kevin Bao	Unsupervised Learning Notebook Unsupervised Learning Final Report Sections: 7, 8, 9.2
Bonseok Koo	Supervised Learning Notebook Supervised Learning Final Report Sections: 5.1, 6, 9.1
Zach Snow	Data Imports and Preprocessing Final Report Sections: 1, 2, 3, 4, 10, assistance on domain-specific knowledge in other sections Organization and formatting of papers and reports

Generative AI note: Claude (Anthropic) was used for assistance in developing the data preprocessing pipeline, including FITS file handling and memory management. For the supervised learning portion, GenAI was used only as coding support for debugging and organizing the notebook workflow. All modeling, analysis decisions, and report writing are the original work of the project team. GenAI was not used in the written report.

References

[1] Yan, R. et al. (2019). MaStar: The MaNGA Stellar Library. *The Astrophysical Journal Supplement Series*, 240(1), 14. https://doi.org/10.3847/1538-4365/aaf5fb

[2] Fabbro, S., Venn, K.A., O'Briain, T., et al. (2018). An Application of Deep Learning in the Analysis of Stellar Spectra. *Monthly Notices of the Royal Astronomical Society*, 475(3), 2978–2993. arXiv:1709.09182.

[3] Anders, F., Khalatyan, A., Queiroz, A., Nepal, S., & Chiappini, C. (2023). Parameters for >300 million Gaia stars: Bayesian inference vs. machine learning. *Highlights of Spanish Astrophysics XI*. arXiv:2302.06995.

[4] Reis, I., Baron, D., & Shahaf, S. (2018). Detecting outliers and learning complex structures with large spectroscopic surveys: a case study with APOGEE stars. *Monthly Notices of the Royal Astronomical Society*, 476(2), 2117–2136. arXiv:1711.00022.

[5] *SIADS 532 Data Mining I, Module 3.3.4 "Principal Component Analysis".*

[6] *SIADS 543 Unsupervised Learning, Week 2, k-means*

Appendix A: Full Feature List

The final feature set consists of 4,614 continuum-normalized flux values per spectrum, indexed by wavelength bin position on the shared MaStar wavelength grid spanning 3,622 to 10,354 Angstroms at R~1800 resolution. Each feature corresponds to a single wavelength bin and carries no derived or engineered information beyond the continuum normalization described in Section 4.3. Bins without at least 95% coverage across the catalog were excluded prior to modeling, reducing the original ~4,614 bins to the final working set. No additional features such as photometry, radial velocity, or stellar labels were used as model inputs for either the supervised or unsupervised components.

Appendix B: Additional Figures and Results

Figure B1: Spectral type distribution across full dataset, training set, and test set

```
Train X: (2632, 4563)
Test X: (659, 4563)

Full spectral type distribution:
SPECTRAL_TYPE
K    0.588271
G    0.192647
F    0.142814
A    0.054391
M    0.021878
Name: proportion, dtype: float64

Train spectral type distribution:
SPECTRAL_TYPE
K    0.588146
G    0.192629
F    0.142857
A    0.054331
M    0.022036
Name: proportion, dtype: float64

Test spectral type distribution:
SPECTRAL_TYPE
K    0.588771
G    0.192716
F    0.142640
A    0.054628
M    0.021244
Name: proportion, dtype: float64
```

Feature Importance: Top 20 Wavelengths by Target

For Teff, the top features cluster tightly around 6561-6569 Å (ranks 1, 2, 3, 5), which is the Hα hydrogen Balmer line at 6562.8 Å. Additional high-importance features appear near Hβ (4855-4866 Å, ranks 6-7) and Hγ (4337 Å, rank 8). Hydrogen Balmer absorption strength is the primary temperature diagnostic across the spectral types in our sample, peaking for A-type stars around 10,000 K and weakening toward both hotter and cooler ends. The feature at 8437 Å (rank 9) falls near the hydrogen Paschen series, a consistent secondary temperature indicator at longer wavelengths.

For log g, the most important features concentrate around Ca II K (3934-3936 Å, ranks 4, 8, 9) and the Mg I b triplet region (5164-5252 Å, ranks 1, 3, 7, 10). Both are well-established surface gravity diagnostics: Ca II and Mg I lines are sensitive to collisional pressure broadening, which scales directly with atmospheric density and therefore with log g. The model independently recovers the same wavelength regions that human spectroscopists use to classify giant versus dwarf stars on the Kiel diagram.

For [Fe/H], the top feature by a significant margin is 5891 Å (rank 1), corresponding to the Na I D doublet at 5889.9/5895.9 Å, one of the strongest metal absorption features in optical spectra. The remaining top features fall in iron-rich line regions between 4800-5200 Å (Fe I blends at ranks 2, 4, 5, 6, 8, 9) and near Mg I b at 5184 Å (rank 10) and Mg II at 4482 Å (rank 3). The model's reliance on iron and alpha-element lines to predict metallicity is a strong sanity check that it is learning real spectral chemistry rather than coincidental correlations.

Table B1: *Top 20 important wavelengths for Teff, log g, and [Fe/H]*

Top wavelengths for INPUT_TEFF

	Target	Rank	Wavelength	Feature_Index	Importance	MAE	R2
0	INPUT_TEFF	1	6561.452637	2581	0.077454	128.543452	0.966865
1	INPUT_TEFF	2	6559.941895	2580	0.075942	128.543452	0.966865
2	INPUT_TEFF	3	6567.498535	2585	0.069688	128.543452	0.966865
3	INPUT_TEFF	4	8757.900391	3835	0.055268	128.543452	0.966865
4	INPUT_TEFF	5	6569.011230	2586	0.047850	128.543452	0.966865
5	INPUT_TEFF	6	4855.120117	1273	0.036931	128.543452	0.966865
6	INPUT_TEFF	7	4864.072266	1281	0.035923	128.543452	0.966865
7	INPUT_TEFF	8	4337.105469	783	0.027521	128.543452	0.966865
8	INPUT_TEFF	9	8437.232422	3673	0.025189	128.543452	0.966865
9	INPUT_TEFF	10	3893.139893	314	0.024014	128.543452	0.966865
10	INPUT_TEFF	11	4860.713379	1278	0.023980	128.543452	0.966865
11	INPUT_TEFF	12	4866.312500	1283	0.023451	128.543452	0.966865
12	INPUT_TEFF	13	4107.711914	547	0.020669	128.543452	0.966865
13	INPUT_TEFF	14	6565.986816	2584	0.019956	128.543452	0.966865
14	INPUT_TEFF	15	4865.192383	1282	0.018460	128.543452	0.966865
15	INPUT_TEFF	16	4870.796875	1287	0.018333	128.543452	0.966865
16	INPUT_TEFF	17	4109.604004	549	0.013036	128.543452	0.966865
17	INPUT_TEFF	18	9225.713867	4061	0.011975	128.543452	0.966865
18	INPUT_TEFF	19	8384.940430	3646	0.010279	128.543452	0.966865
19	INPUT_TEFF	20	4343.101562	789	0.009846	128.543452	0.966865

Top wavelengths for INPUT_LOGG

	Target	Rank	Wavelength	Feature_Index	Importance	MAE	R2
0	INPUT_LOGG	1	5251.701172	1614	0.057747	0.268047	0.887364
1	INPUT_LOGG	2	4410.623535	856	0.044213	0.268047	0.887364
2	INPUT_LOGG	3	5250.492188	1613	0.038772	0.268047	0.887364
3	INPUT_LOGG	4	3936.406982	362	0.035345	0.268047	0.887364
4	INPUT_LOGG	5	4205.329102	649	0.029313	0.268047	0.887364
5	INPUT_LOGG	6	4771.995117	1198	0.025774	0.268047	0.887364
6	INPUT_LOGG	7	5164.163574	1541	0.024871	0.268047	0.887364
7	INPUT_LOGG	8	3935.500732	361	0.022390	0.268047	0.887364
8	INPUT_LOGG	9	3934.594727	360	0.018246	0.268047	0.887364
9	INPUT_LOGG	10	5165.353027	1542	0.017023	0.268047	0.887364
10	INPUT_LOGG	11	4457.588867	902	0.016108	0.268047	0.887364
11	INPUT_LOGG	12	3938.220215	364	0.016032	0.268047	0.887364
12	INPUT_LOGG	13	5254.120117	1616	0.015656	0.268047	0.887364
13	INPUT_LOGG	14	5252.910645	1615	0.015533	0.268047	0.887364
14	INPUT_LOGG	15	8184.647949	3541	0.014077	0.268047	0.887364
15	INPUT_LOGG	16	4957.925781	1364	0.013489	0.268047	0.887364
16	INPUT_LOGG	17	5615.649414	1905	0.011984	0.268047	0.887364
17	INPUT_LOGG	18	6558.431641	2579	0.009984	0.268047	0.887364
18	INPUT_LOGG	19	3937.313477	363	0.009559	0.268047	0.887364
19	INPUT_LOGG	20	3887.764893	308	0.009389	0.268047	0.887364

Top wavelengths for INPUT_FE_H

	Target	Rank	Wavelength	Feature_Index	Importance	MAE	R2
0	INPUT_FE_H	1	5891.148926	2113	0.082836	0.170777	0.864083
1	INPUT_FE_H	2	4873.040039	1289	0.023069	0.170777	0.864083
2	INPUT_FE_H	3	4482.290527	926	0.022756	0.170777	0.864083
3	INPUT_FE_H	4	4874.162598	1290	0.021137	0.170777	0.864083
4	INPUT_FE_H	5	5101.524902	1488	0.020442	0.170777	0.864083
5	INPUT_FE_H	6	4667.668457	1102	0.020190	0.170777	0.864083
6	INPUT_FE_H	7	9259.765625	4077	0.019514	0.170777	0.864083
7	INPUT_FE_H	8	5003.800781	1404	0.015768	0.170777	0.864083
8	INPUT_FE_H	9	4985.399902	1388	0.014998	0.170777	0.864083
9	INPUT_FE_H	10	5184.417969	1558	0.014407	0.170777	0.864083
10	INPUT_FE_H	11	8667.623047	3790	0.014281	0.170777	0.864083
11	INPUT_FE_H	12	5277.156738	1635	0.013791	0.170777	0.864083
12	INPUT_FE_H	13	8548.698242	3730	0.013136	0.170777	0.864083
13	INPUT_FE_H	14	5186.806152	1560	0.011599	0.170777	0.864083
14	INPUT_FE_H	15	7166.382812	2964	0.011181	0.170777	0.864083
15	INPUT_FE_H	16	5004.953125	1405	0.009065	0.170777	0.864083
16	INPUT_FE_H	17	9063.584961	3984	0.008984	0.170777	0.864083
17	INPUT_FE_H	18	7837.905273	3353	0.008585	0.170777	0.864083
18	INPUT_FE_H	19	4666.593750	1101	0.008352	0.170777	0.864083
19	INPUT_FE_H	20	9011.560547	3959	0.007555	0.170777	0.864083

Figures B2-B3 show XGBoost performance across all three stellar parameters as a function of n_estimators (50-900) and max_depth (2-8). The Teff curves are discussed in the main text; log g and [Fe/H] follow the same pattern with slightly more variance across folds.

Figure B2: *XGBoost Sensitivity to number of estimators*

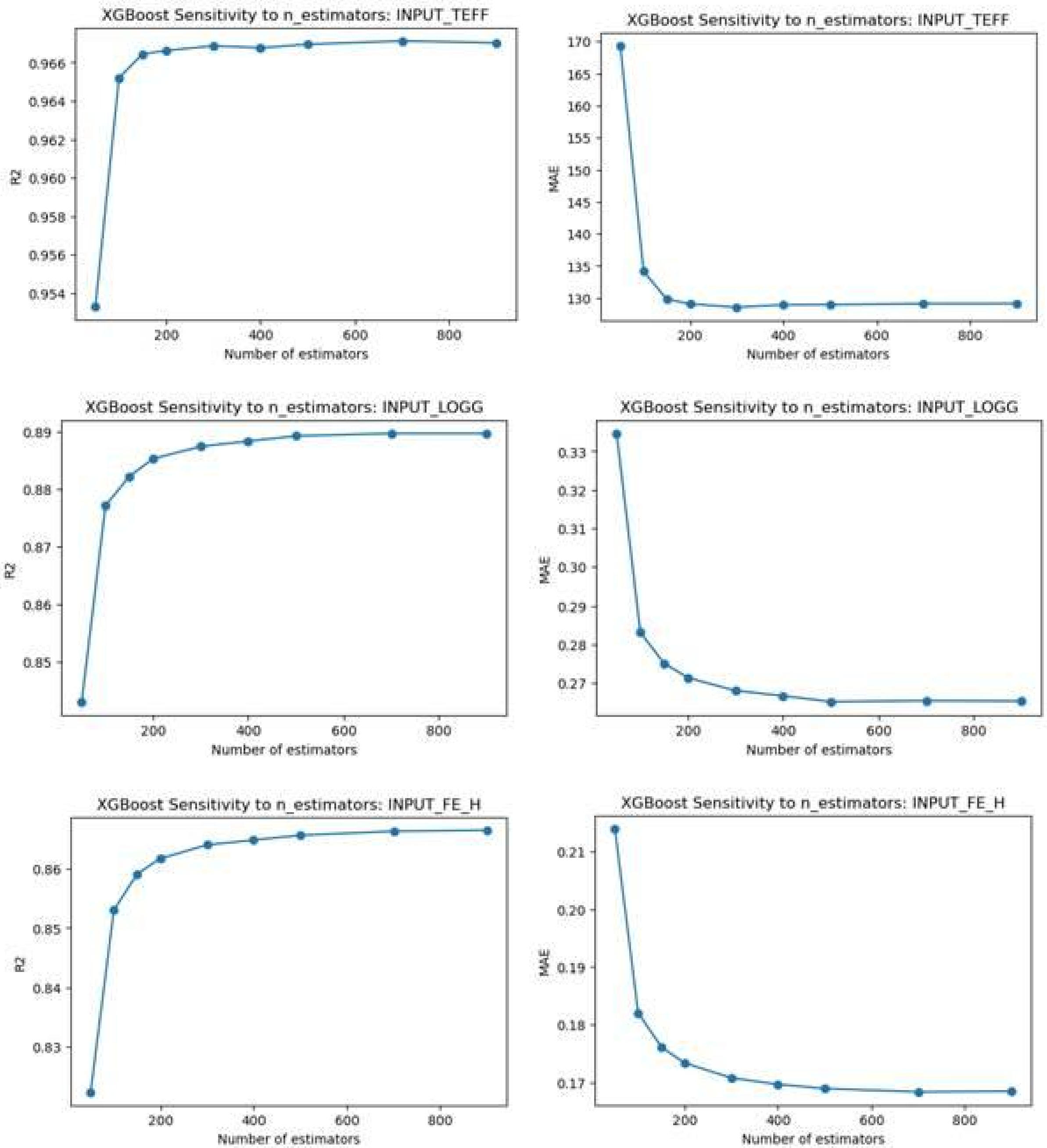


Figure B3: *XGBoost Sensitivity to maximum tree depth*

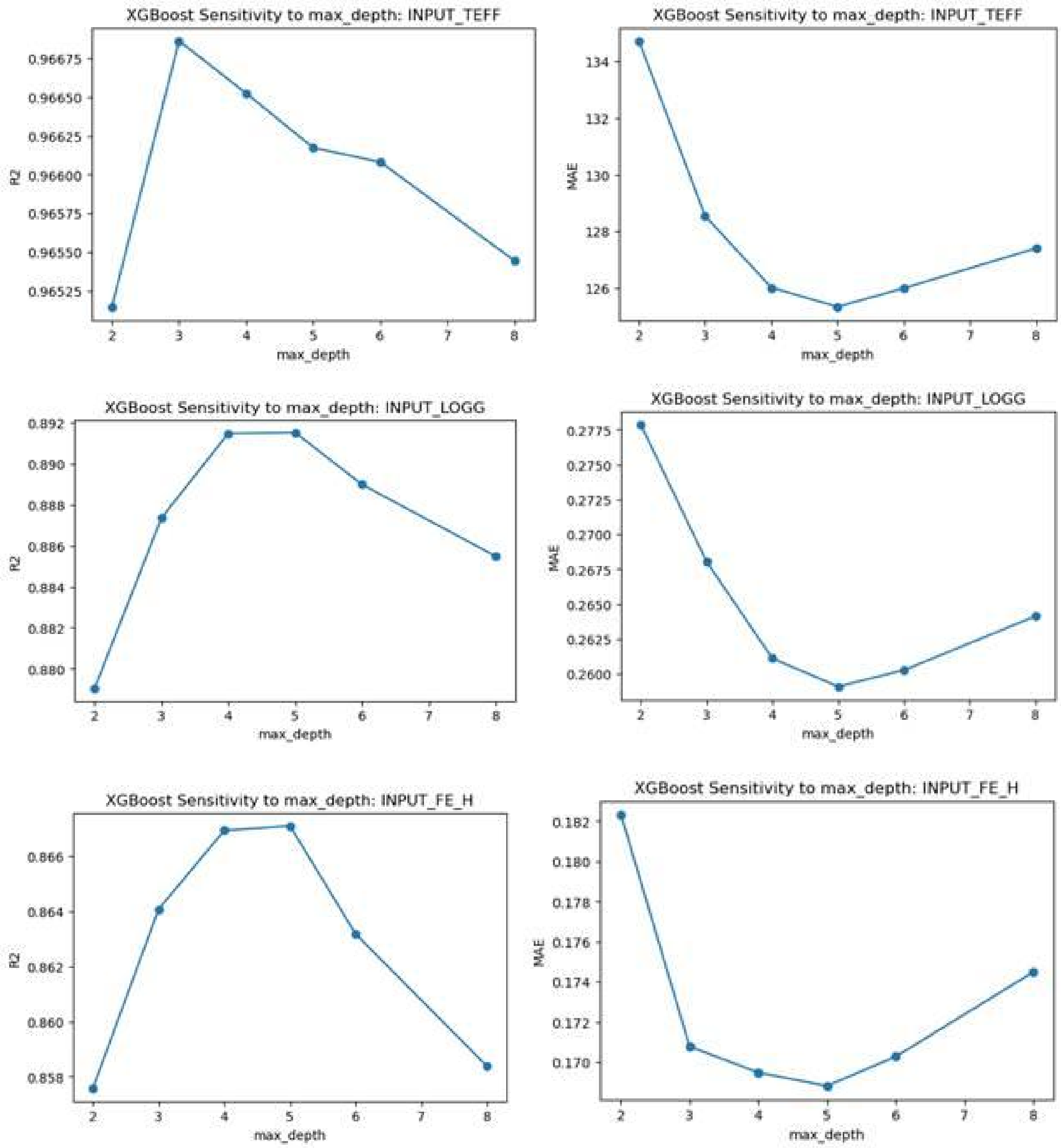


Figure B4: *HDBSCAN hyperparameter grid sweep results across min_cluster_size and min_samples combinations, showing n_clusters, noise fraction, DBCV, and persistence.*

```
HDBSCAN hyperparameter sweep:
 min_cluster_size  min_samples  n_clusters  noise_frac   dbcv  persistence
              100            5           2       0.493 0.0001       0.2231
              100           10           2       0.470 0.0000       0.2275
              100           25           2       0.346 0.0001       0.2826
              200            5           2       0.493 0.0001       0.2185
              200           10           2       0.470 0.0000       0.2282
              200           25           2       0.346 0.0001       0.2860
              400            5           0       1.000 0.0000       0.0000
              400           10           0       1.000 0.0000       0.0000
              400           25           2       0.346 0.0001       0.2722
Note: DBCV is low/flat across the grid -> the data show no strong density-separated structure

Selected by persistence within noise band (0.1, 0.6): min_cluster_size=200, min_samples=25 (DBCV=0.0001, persistence=0.286, noise=34.6%)

HDBSCAN found 2 clusters; 34.6% of stars labeled noise.
HDBSCAN cluster sizes:
-1    5266
 0     549
 1    9385
```

Table B5: *K-means cluster stability ablation results. Left: silhouette score and ARI across N_PCA dimensions (10, 30, 50, 100). Right: ARI after dropping the Ca H&K blue-end bins below 4200 Å.*

```
ABLATION 1 - cluster stability vs PCA dimensionality:
 pca_dims  silhouette  ARI_vs_pca50
       10       0.594           1.0
       30       0.558           1.0
       50       0.551           1.0
      100       0.544           1.0

ABLATION 2 - drop the blue Ca H&K block (<4200 A) and re-cluster:
Dropped 610 blue bins (<4200 A, incl. Ca H&K).
ARI(full vs no-blue) = 0.981
```

Figure B6: *Kiel diagram (Teff vs. log g) for the 15,200-star unsupervised dataset colored by K-means cluster assignment at K=2, showing the temperature-driven split between warm (cluster 1) and cool (cluster 0) stellar populations.*

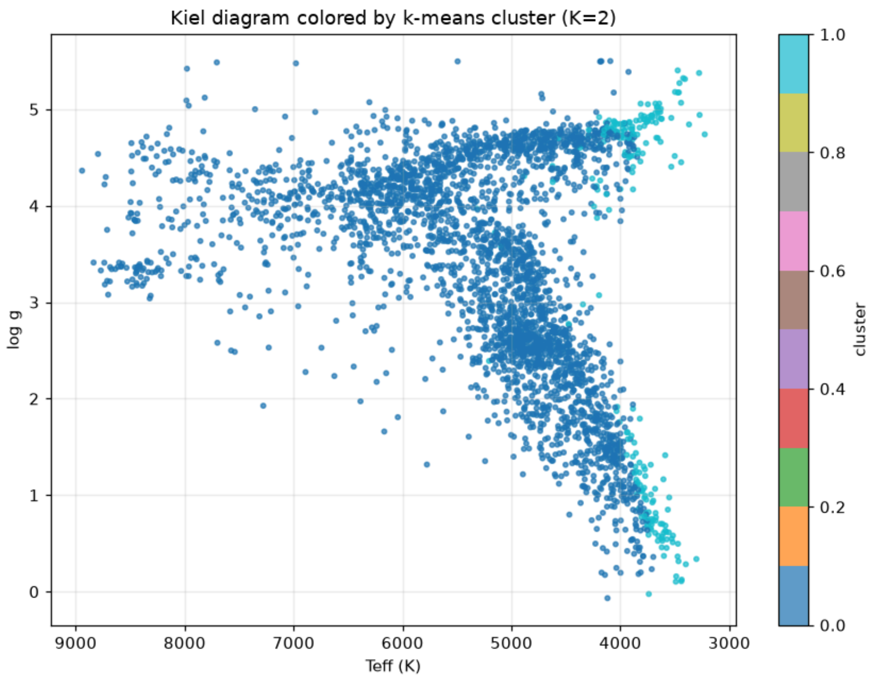


Figure B7: *Kiel diagram colored by K-means cluster assignment at K=6, compared against known spectral type labels as a post-hoc sanity check. Label data was not used in modeling.*

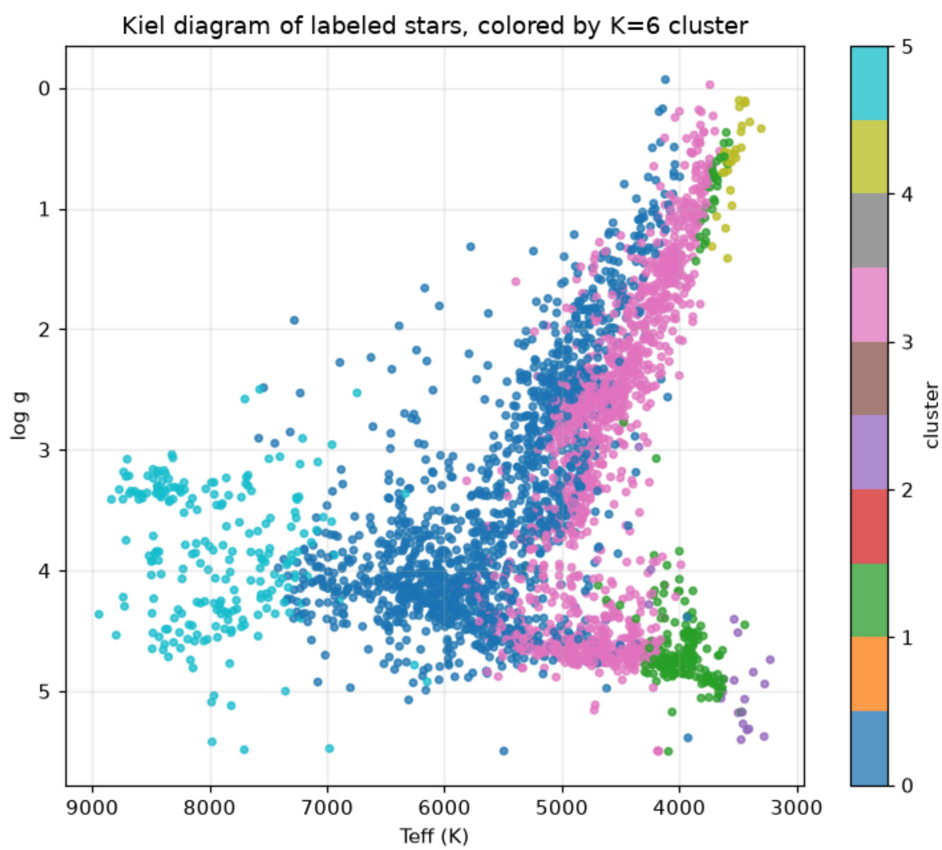
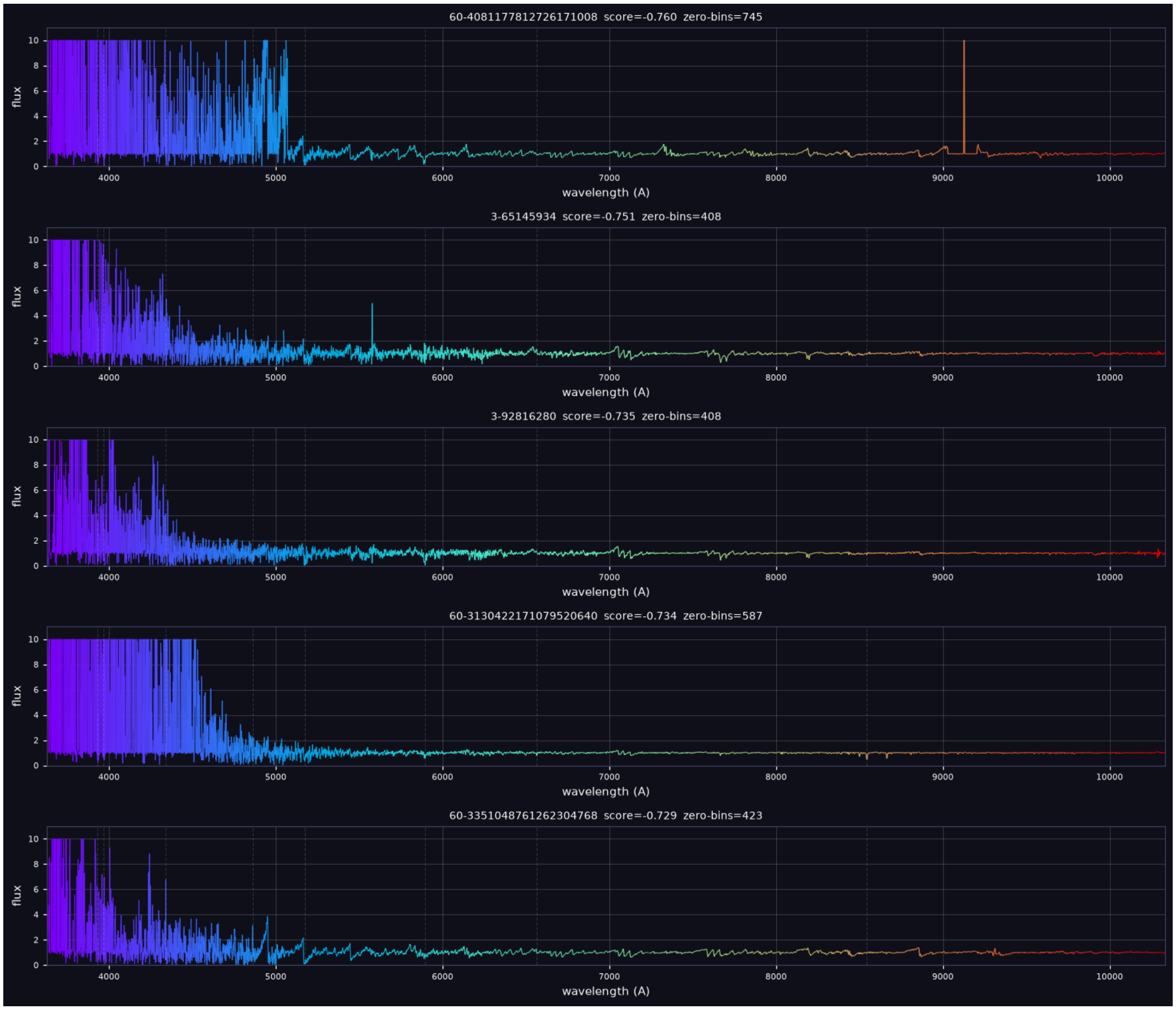


Figure B8: *Five highest-ranked Isolation Forest anomalies by anomaly score.*



Appendix C: Code and Preprocessing Notes

All project code is available at: https://github.com/kevin7bao/mastar-stellar-ml

Repository structure: Notebooks are organized by pipeline stage and should be run in order: preprocessing -> EDA -> supervised -> unsupervised. All notebooks are written for Google Colab and mount Google Drive for data access.

Preprocessing outputs: The pipeline produces five files written to Google Drive:

- mastar_labeled_meta.csv (3,291 × 5) - metadata and stellar parameter labels for supervised learning
- mastar_labeled_spectra.npy (3,291 × 4,563) - continuum-normalized flux arrays for labeled stars
- mastar_full_meta.csv (15,200 rows) - metadata for all quality-filtered stars
- mastar_full_spectra.npy (15,200 × 4,563) - continuum-normalized flux arrays for all stars
- mastar_wave.npy (4,563) - shared wavelength grid in Angstroms

Data access: Raw FITS files (~5 GB compressed) are downloaded directly from SDSS DR17 within the preprocessing notebook and are not committed to the repository. A 100-record sample (data/samples/mastar_sample_100.csv) is included in the repo to satisfy the course data submission requirement. Full processed outputs are stored in the team's shared Google Drive.

Dependencies: numpy, pandas, scikit-learn, matplotlib, seaborn, astropy, scipy, tqdm.